4 ANALYSE THE MALICIOUS NETWORK TRAFFIC USING WIRESHARK

AIM

To analyze malicious network traffic, investigate command-and-control (C2) activity, and identify indicators of compromise using packet capture analysis in the TryHackMe room “Carnage.”.

PROCEDURE

1. Open the TryHackMe room “Carnage” and load the provided PCAP file in Wireshark.
2. Study the suspicious HTTP, DNS, and SMTP traffic to identify malicious connections.
3. Analyze downloaded files, domains, and Cobalt Strike C2 server activity.
4. Complete the tasks and answer the questions based on the traffic investigation.

QUESTIONS

TASK 2 – Traffic Analysis

What are the two IP addresses of the Cobalt Strike servers? Use VirusTotal (the Community tab) to confirm if IPs are identified as Cobalt Strike C2 servers. (answer format: enter the IP addresses in sequential order)

185.106.96.158, 185.125.204.174 — ✓ Correct Answer

What is the Host header for the first Cobalt Strike IP address from the previous question?

ocsp.verisign.com — ✓ Correct Answer

What is the domain name for the first IP address of the Cobalt Strike server? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

survmeter.live — ✓ Correct Answer

What is the domain name of the second Cobalt Strike server IP? You may use VirusTotal to confirm if it's the Cobalt Strike server (check the Community tab).

securitybusinpuff.com — ✓ Correct Answer

What is the domain name of the post-infection traffic?

maldivehost.net — ✓ Correct Answer

What are the first eleven characters that the victim host sends out to the malicious domain involved in the post-infection traffic?

zLIisQRWZI9 — ✓ Correct Answer

What was the length for the first packet sent out to the C2 server?

281 — ✓ Correct Answer

What was the Server header for the malicious domain from the previous question?

Apache/2.4.49 (cPanel) OpenSSL/1.1.1l mod_bwlimited/1.4 — ✓ Correct Answer

The malware used an API to check for the IP address of the victim's machine. What was the date and time when the DNS query for the IP check domain occurred? (**answer format**: yyyy-mm-dd hh:mm:ss UTC)

2021-09-24 17:00:04 — ✓ Correct Answer

What was the domain in the DNS query from the previous question?

api.ipify.org — ✓ Correct Answer

Looks like there was some malicious spam (malspam) activity going on. What was the first MAIL FROM address observed in the traffic?

farshin@mailfa.com — ✓ Correct Answer

How many packets were observed for the SMTP traffic?

1439 — ✓ Correct Answer

RESULT

Successfully learned how to investigate malicious traffic, detect C2 communication, and identify indicators of compromise using Wireshark and threat analysis techniques.